

JKBOSE 2026

CLASS 11TH

**PERMUTATION AND
COMBINATIONS**

MATHS

LECTURE-04

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TOPICS TO BE COVERED

1 **Permutation (when all objects are not distinct)**

2 **Exercise Discussion 6.3**

3 **Related Problems**

Permutations When all the objects are not distinct:

* Consider the letters : A, A, B.

Question: How many permutations are possible by taking all at a time?

Ans) Possible permutations are:
AAB, ABA, BAA

$\Rightarrow$ No. of permutations = 3

$$\Rightarrow \text{No. of permutations} = \frac{3!}{2!}$$

* Consider the letters : A, A, B, B.

Question: How many arrangements (permutations) are possible by taking all at a time?

Ans) Possible permutations are:

AABB, ABAB, ABBA, BABA, BAAB, BBAA

$$\Rightarrow \text{No. of permutations} = 6 = \frac{4!}{2!2!}$$

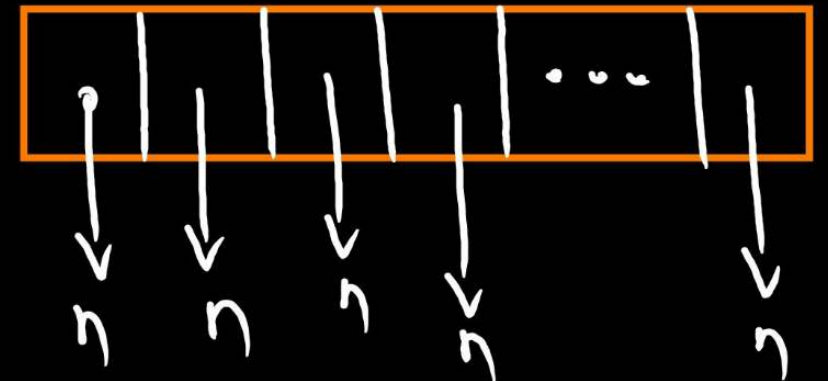
$$\Rightarrow \text{No. of permutations} = \frac{4!}{2!2!}$$

* No. of Permutations of n objects in which p objects are identical is $\frac{n!}{p!}$

* No. of Permutations of n objects in which p_1 objects are of one kind, p_2 objects are of second ^{kind}, ... p_k objects are of k^{th} kind, is $\frac{n!}{p_1! p_2! \dots p_k!}$

* No. of Permutations of n -objects by taking r at a time in which the repetition is allowed is n^r .

1, 2, ..., n



$$n \times n \times n \times \dots \times n \\ = n^r$$

EXAMPLE

Find the number of different permutations of the letters of the word, 'BANANA'.

(Sol) Required No. of Permutations

$$= \frac{6!}{3!2!} = \frac{6 \times 5 \times \cancel{4} \times \cancel{3} \times \cancel{2} \times \cancel{1}}{\cancel{3}! \times \cancel{2} \times \cancel{1}}$$

$$= 6 \times 10$$

$$= 60 \quad \text{Hence the result.}$$

EXAMPLE

Find the number of different 8-letter arrangements that can be made from the letters of the word **DAUGHTER** so that

(i) All vowels occur together

(ii) No two vowels are occur together

(Sol) (i) Required Number of Permutations

$$= 6! \times 3!$$

$$= 720 \times 6$$

$$= 4320$$

AUEDGHTR

D*G*H*T*R

(ii) Required No. of Permutations

$$= 5! \times 6P_3$$

$$= 120 \times \frac{6!}{3!} = 120 \times \frac{6 \times 5!}{6}$$

$$= 120 \times 120$$

$$= 14400$$

Hence the result.

EXAMPLE

IEEEENDPNDNC



Find the number of arrangements of the letters of the word INDEPENDENCE. In how many of these arrangements,

- (i) Do the words start with P
- (ii) Do all the vowels always occur together
- (iii) Do the vowels never occur together
- (iv) Do the words begin with I and end in P?

(i) (Sol) Required No. of Permutations

$$= 1 \times \frac{11!}{3! 4! 2!}$$
$$= \frac{11!}{3! 4! 2!} = \underline{\hspace{2cm}}$$

(ii) (Sol) Required Number of Permutations

$$= \frac{8!}{3! 2!} \times \frac{5!}{4!} = \underline{\hspace{2cm}}$$

(iii) (iv) H/w.

QUESTION (JEE-MAINS 2021)

The number of arrangements of the letters of the word "INDEPENDENCE" in which all the vowels always occur together is

1 ✓ 16800

2 ✗ 33600

3 ✗ 18000

4 ✗ 14800

(Sol) Required No. of Permutations

$$= \frac{8!}{3!2!} \times \frac{5!}{4!2!}$$

$$= \frac{8 \times 7 \times 6 \times 5 \times \cancel{4} \times 3!}{\cancel{3!} \times \cancel{2}} \times \frac{5 \times \cancel{4}!}{\cancel{4!}}$$

$$= 16800.$$

QUESTION (NCERT EXAMPLAR)

All the letters of the word 'EAMCOT' are arranged in different possible ways. The number of such arrangements in which no two vowels are adjacent to each other is

1

360 ✗

2

144 ✓

3

72 ✗

4

54 ✗

(Sol) Required No. of Permutations

$$= 3! \times {}^4P_3$$

$$= 3! \times 4! = 6 \times 24$$

$$\overline{(4-3)!} = 144.$$

A E O

*M*C*T*

QUESTION (JEE MAINS 2024)

The total number of three digit numbers, divisible by 3, which can be formed by using the digits 1, 3, 5, 8, if repetition of digits is allowed, is:

1 22

2 18

3 21

4 20

Homework.

QUESTION (JEE MAINS 2023)

The letters of the word “**UGHT**” are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word “TOUGH”.

Homework.

1 89

2 84

3 86

4 79

*Dictionary order problem
/ lexicographic
order problem.*

EXERCISE – 6.3

QUESTION

How many 3-digit numbers can be formed by using the digits 1 to 9 if no digit is repeated?

1, 2, 3, 4, 5, 6, 7, 8, 9

$${}^9P_3 \#$$

nP_r

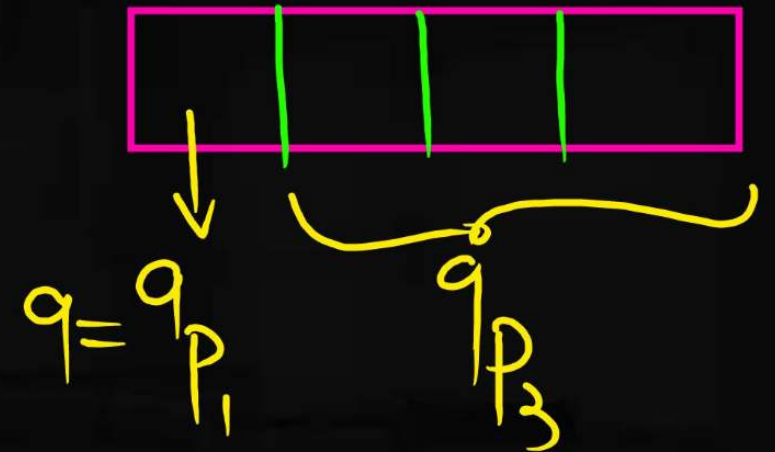
QUESTION

How many 4-digit numbers are there with no digit repeated?

(Sol) Required No. of Permutations

$${}^9P_1 \times {}^9P_3 = 9 \times \frac{9!}{6!} = ??$$

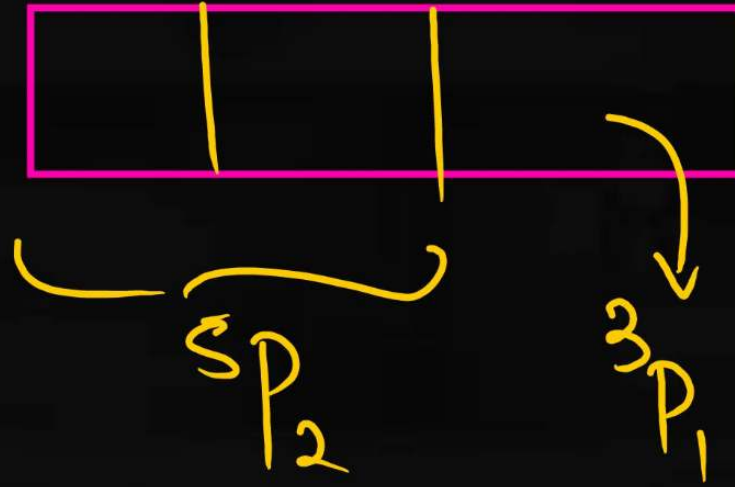
0, 1, 2, 3, 4, 5, 6, 7, 8, 9



QUESTION

How many 3-digit even numbers can be made digits 1, 2, 3, 4, 6, 7, if no digit is repeated?

$${}^5P_2 \times {}^3P_1$$



QUESTION

Find the number of 4-digit numbers that can be formed using the digits 1, 2, 3, 4, 5 if no digit is repeated. How many of these will be even?

Homework.

QUESTION

From a committee of 8 persons, in how many ways can we choose a chairman and a vice chairman assuming one person cannot hold more than one position?

1p/w

8P_2 ✓

8	7
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$$\frac{8!}{6!} = \frac{8 \times 7 \times \cancel{6!}}{\cancel{6!}}$$

$$= 56$$

QUESTION

Find n if ${}^{n-1}P_3 : {}^nP_4 = 1 : 9$

(Sol) Given that ${}^{n-1}P_3 : {}^nP_4 = 1 : 9$

$$\Rightarrow \frac{(n-1)!}{(n-1-3)!} : \frac{n!}{(n-4)!} = 1 : 9$$

$$\Rightarrow \frac{\cancel{(n-1)!}}{\cancel{(n-4)!}} \times \frac{\cancel{(n-4)}}{n \cancel{(n-1)!}} = \frac{1}{9}$$

$$\Rightarrow \frac{1}{n} = \frac{1}{9} \Rightarrow n = 9$$

Hence the result.

QUESTION

Find r if

(i) ${}^5P_r = 2 {}^6P_{r-1}$

(ii) ${}^5P_r = {}^6P_{r-1}$ (H/W)

(Sol) (i) Given that ${}_5P_r = 2 \times {}_6P_{r-1}$

$$\Rightarrow \frac{5!}{(5-r)!} = 2 \times \frac{6!}{(6-r+1)!}$$

$$\Rightarrow \frac{\cancel{5!}}{(5-r)!} = 2 \times 6 \times \frac{\cancel{5!}}{(7-r)!}$$

$${}_n P_r, 0 < r \leq n$$



$$\Rightarrow \frac{(7-r)!}{(5-r)!} = 12$$

$$\Rightarrow \frac{(7-r)(6-r)\cancel{(5-r)!}}{\cancel{(5-r)!}} = 12$$

$$\Rightarrow 42 - 7r - 6r + r^2 - 12 = 0$$

$$\Rightarrow r^2 - 13r + 30 = 0$$

$$\Rightarrow (r-10)(r-3) = 0$$

$$\Rightarrow r = 10, 3$$

$$\Rightarrow r = 3$$

Hence the result.

QUESTION

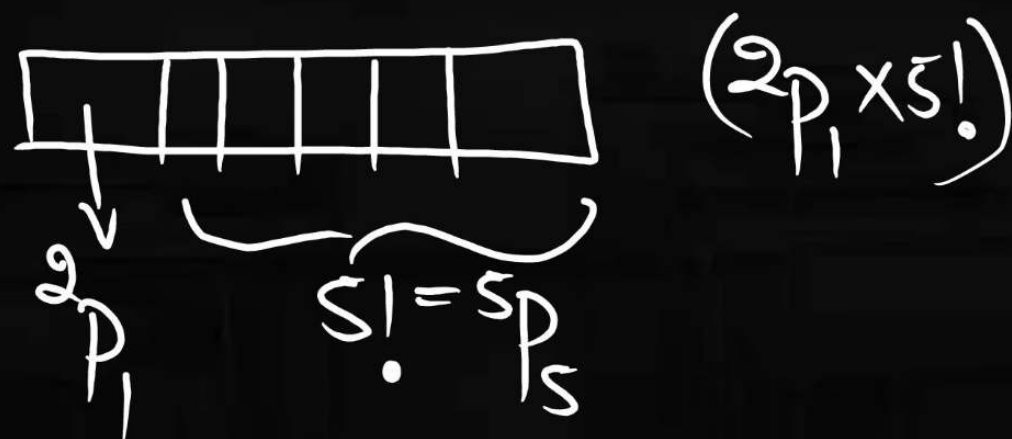
How many words, with or without meaning, can be formed using all the letters of the word **EQUATION** using each letter exactly once?

8! ✓

QUESTION

How many words, with or without meaning can be made from the letters of the word **MONDAY**, assuming that no letter is repeated, if.

- (i) 4 letters are used at a time, $({}^6P_4)$
- (ii) All letters are used at a time, $(6!)$
- (iii) All letters are used but first letter is a vowel?



QUESTION



In how many of the distinct permutations of the letters in **MISSISSIPPI** do the **four I's** **not come together**?

(Sol) Total No. of Permutations

$$= \frac{11!}{4!4!2!}$$

No. of Permutations in which all the four I's are Together

$$= \frac{8!}{4!2!} \times \frac{4!}{4!} = \frac{8!}{4!2!}$$

So, the Required No. of Permutations

$$= \frac{11!}{4!4!2!} - \frac{8!}{4!2!} = -$$

IIIIMISSSSPP

Hence the result



Thank
You